

Main PCB Schematic

GLD V4 (MotherBoardGLDVer2) - section 6(a).7

Item	Details
Type	Actual electrical schematic (not a block diagram), cut into functional blocks from the original EasyEDAfile. Every component has a numbered boundary box; the same number is used in the description table.
Source	GLD ATEX.zip > 1-Schematic_MotherBoardGLDVer2.json (EasyEDA6.5.57)
Source revision / date	1.0 / 2026-07-19 (from the title block of the EasyEDAschematic)
Document number	PGLD-GLD-V4-MB-SCH-NN (proposed; NN = figure number)
Number of figures	26

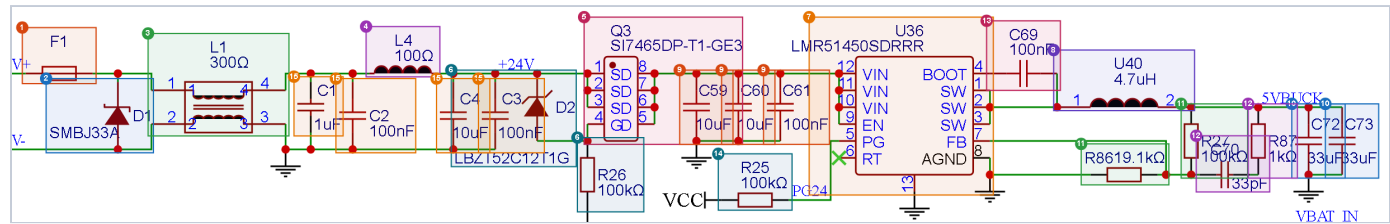
Coverage of the requirements

Required content	Figures / sections
24 VDC input	Figures 1, 2
Input protection	Figures 3; Figures 1, 2 (F1, D1, L1, L4, F2, Q2) and the protection table
5 V rail	Figures 4, 5, 6; Figure 1 (buck U36)
3.3 V rail	Figures 7, 8, 9, 10, 11
MQ heater supply	Figures 12, 13, 14
MQ sensor bridge / analog circuitry	Figures 15, 16, 17, 18; AIN input filters in Figure 19
ADC	Figures 19, 20
ESP32-S3	Figures 21, 22
LoRa	Figure 23
Fan driver	Figure 24
LED / buzzer driver	Figure 25; status LED in Figure 21
RS-485 / Modbus	Figure 26
Protection devices	Section "Protection devices" (table)
Grounding	Section "Grounding"

Every figure carries a sheet information table: drawing/document number, revision, date, title and sheet number.

24 VDC input

Figure 1 - 24 VDC input, input protection and 24 V to 5 V buck (U36)

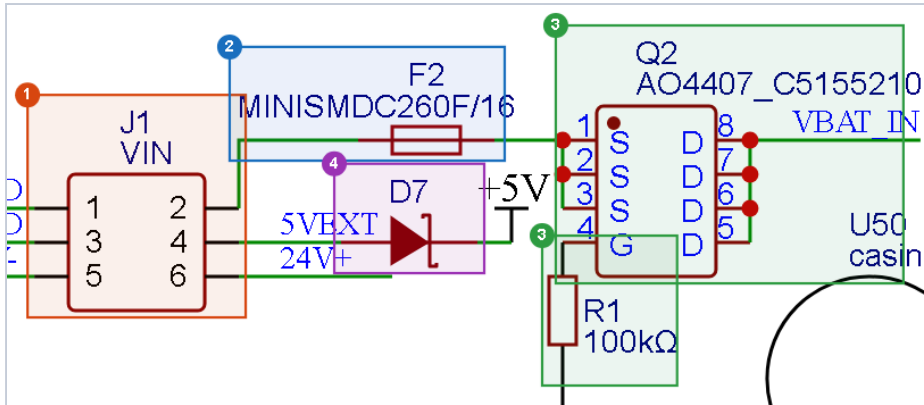


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-01	1.0	2026-07-19	24 VDC input, input protection and 24 V to 5 V buck (U36)	1 of 26

Block function: The 24 VDC path from the connector to the 24 V to 5 V buck (U36). **Requirement covered:** 24 VDC input and input protection

No	Component	What it is	What it does
1	F1	Resettable PTC fuse MINISMD260F/16-2 (2.6 Ahold).	Opens on a short circuit or overload on the 24 V line and recovers by itself. Note: the /16 suffix in the part number usually means a 16 V voltage rating - to be confirmed against the datasheet because this line is 24 V.
2	D1	TVS diode SMBJ33A(33 V standoff).	Clips voltage surges/transients between the 24 V line after F1 and 24V-, so the circuitry behind it is not damaged.
3	L1	Common-mode choke ACM7060-301-2PL (4 pins).	Suppresses common-mode noise on the 24V+ / 24V- pair. Winding 1-4 carries 24V+, winding 2-3 carries the return side (24V- to GND).
4	L4	Ferrite bead MPZ2012S101A(100 ohm @ 100 MHz).	High-frequency noise filter in series with the +24V line before it enters the internal rail.
5	Q3	P-channel MOSFET SI7465DP (source on +24V, drain toward the buck input).	Passes +24V to the 5 V converter. Its source is on the +24V side and its drain on the load side; this orientation should be verified if the part is meant to provide reverse-polarity protection.
6	D2, R26	12 V zener (LBZT52C12) and a 100k resistor to GND.	R26 pulls the gate to GND so Q3 turns on. D2 limits the gate-source voltage of Q3 to 12 V so the gate is not damaged by 24 V.
7	U36	Synchronous buck IC LMR51450 (VIN up to 36 V).	Step-down converter from 24 V to 5VBUCK. EN is tied to VIN (always on); the PG (power good) pin goes to the ESP32 as signal PG24.
8	U40	4.7 uH inductor (GSSM06304R7M2AU).	Buck energy-storage inductor between the SW node and 5VBUCK.
9	C59, C60, C61	Input capacitors 10 uF/50 V x2 + 100 nF.	Supply the pulsed input current of the buck.
10	C72, C73	Output capacitors 33 uF x2 + 10 uF.	Smooth the 5VBUCK output voltage.
11	R27, R86	FB voltage divider 100k / 19.1k.	Sets the output: $0.8\text{ V} \times (1 + 100/19.1) = \text{about } 5.0\text{ V}$ (calculated from the resistor values).
12	R87, C70	1k resistor in series with a 33 pF capacitor on FB.	Feed-forward network for stability / transient response.
13	C69	100 nF BOOT-SW capacitor.	Bootstrap for the internal high-side MOSFET driver.
14	R25	100k pull-up to VCC on PG24.	Gives a 3.3 V logic level for the power-good signal read by the ESP32 (IO47).
15	C1, C2, C3, C4	Decoupling capacitors 1 uF, 100 nF, 100 nF, 10 uF/50 V.	Filter ripple/noise and hold up small transients on the +24V rail (paired with L1/L4).

Figure 2 - Power connector J1, battery input and external 5 V



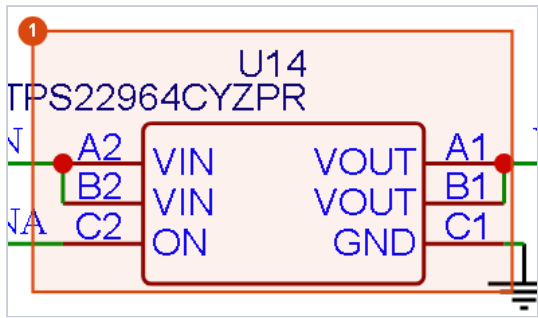
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-02	1.0	2026-07-19	Power connector J1, battery input and external 5 V	2 of 26

Block function: Power connector, battery input protection and switch, and the external 5 V path. **Requirement covered:** 24 VDC input and input protection

No	Component	What it is	What it does
1	J1	Power connector, 2x3 pins (J1).	The only power entry from outside: pin 6 = 24V+, pin 5 = 24V-, pin 4 = 5VEXT (external 5 V), pins 3 and 1 = GND, pin 2 = battery input (goes to F2).
2	F2	PTC fuse MINISMDC260F/16-2.	Overcurrent protection for the battery input at J1 pin 2.
3	Q2, R1	P-MOSFET AO4407 with a 100k gate resistor to GND.	Passes the battery to net VBAT_IN. R1 turns Q2 on through the gate to GND. The source-on-battery orientation should be verified if the part is meant to provide reverse-polarity protection.
4	D7	SS54 Schottky diode from 5VEXT to +5V.	Lets external 5 V (J1 pin 4) supply the +5V rail without back-feeding.

Input protection

Figure 3 - Battery load switch (U14)



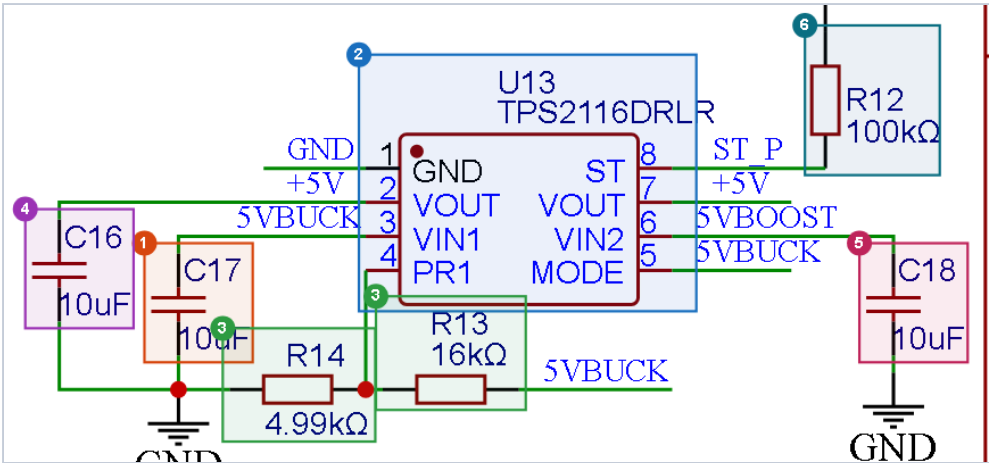
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-03	1.0	2026-07-19	Battery load switch (U14)	3 of 26

Block function: Battery load switch to the VBAT rail. **Requirement covered:** Input protection

No	Component	What it is	What it does
1	U14	Load switch TPS22964C (VIN = VBAT_IN, VOUT = VBAT, ON = ENA).	Connects/disconnects the battery to the whole VBAT rail (boost converters) according to the ENAsignal from the power-latch circuit.

5 V rail

Figure 4 - 5 V power-path mux (U13)

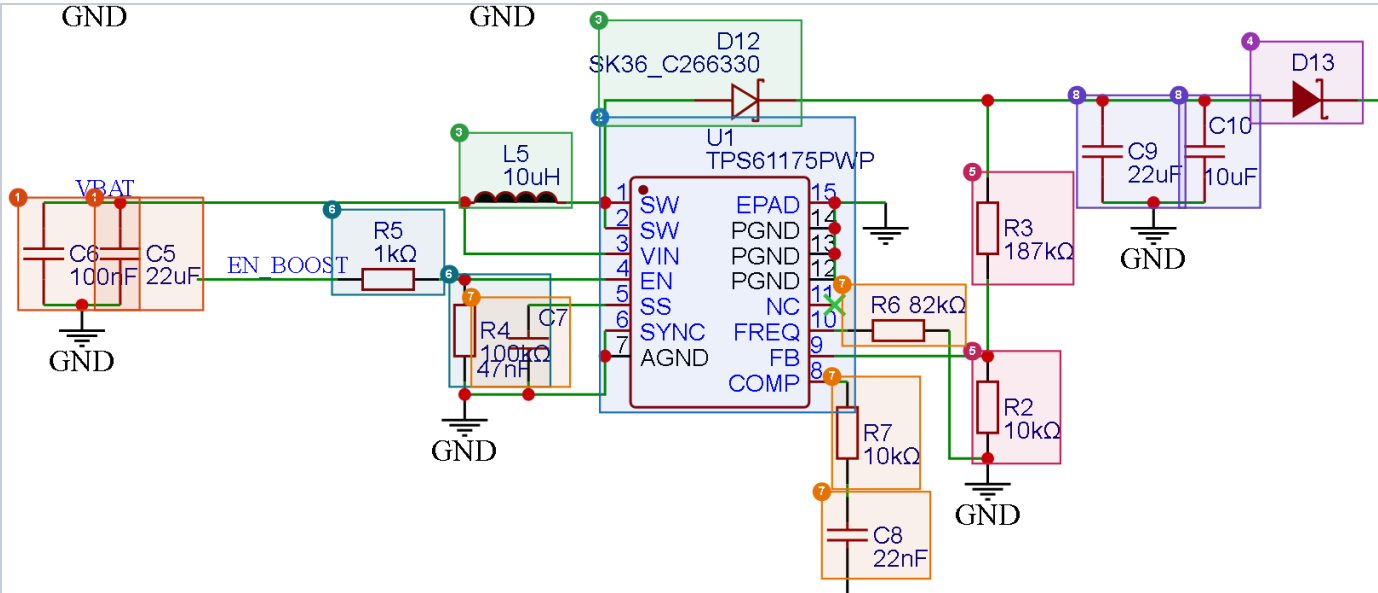


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-04	1.0	2026-07-19	5 V power-path mux (U13)	4 of 26

Block function: Selects the 5 V source between 5VBUCK and 5VBOOST. **Requirement covered:** 5 V rail

No	Component	What it is	What it does
1	C17	Output capacitors 33 uF x2 + 10 uF.	Smooth the 5VBUCK output voltage.
2	U13	Power mux TPS2116 (VIN1 = 5VBUCK, VIN2 = 5VBOOST, VOUT = +5V).	Automatically selects the 5 V source with priority on 5VBUCK (MODE is tied to 5VBUCK) and switches to 5VBOOST when 5VBUCK is lost. The ST pin goes to the ESP32 as ST_P (source status).
3	R13, R14	16k / 4.99k divider on PR1.	Sets the voltage threshold at which 5VBUCK is considered valid before the mux gives it priority.
4	C16	10 uF and 100 nF capacitors on +5V.	Decoupling of the +5V rail.
5	C18	VBAT input and 5VBOOST output capacitors.	Decoupling on the input and output sides of the boost.
6	R12	100k pull-up to VCC on ST_P.	Gives a 3.3 V logic level for the mux status signal read by the ESP32 (IO18).

Figure 5 - Battery boost to about 24 V (U1, TPS61175)

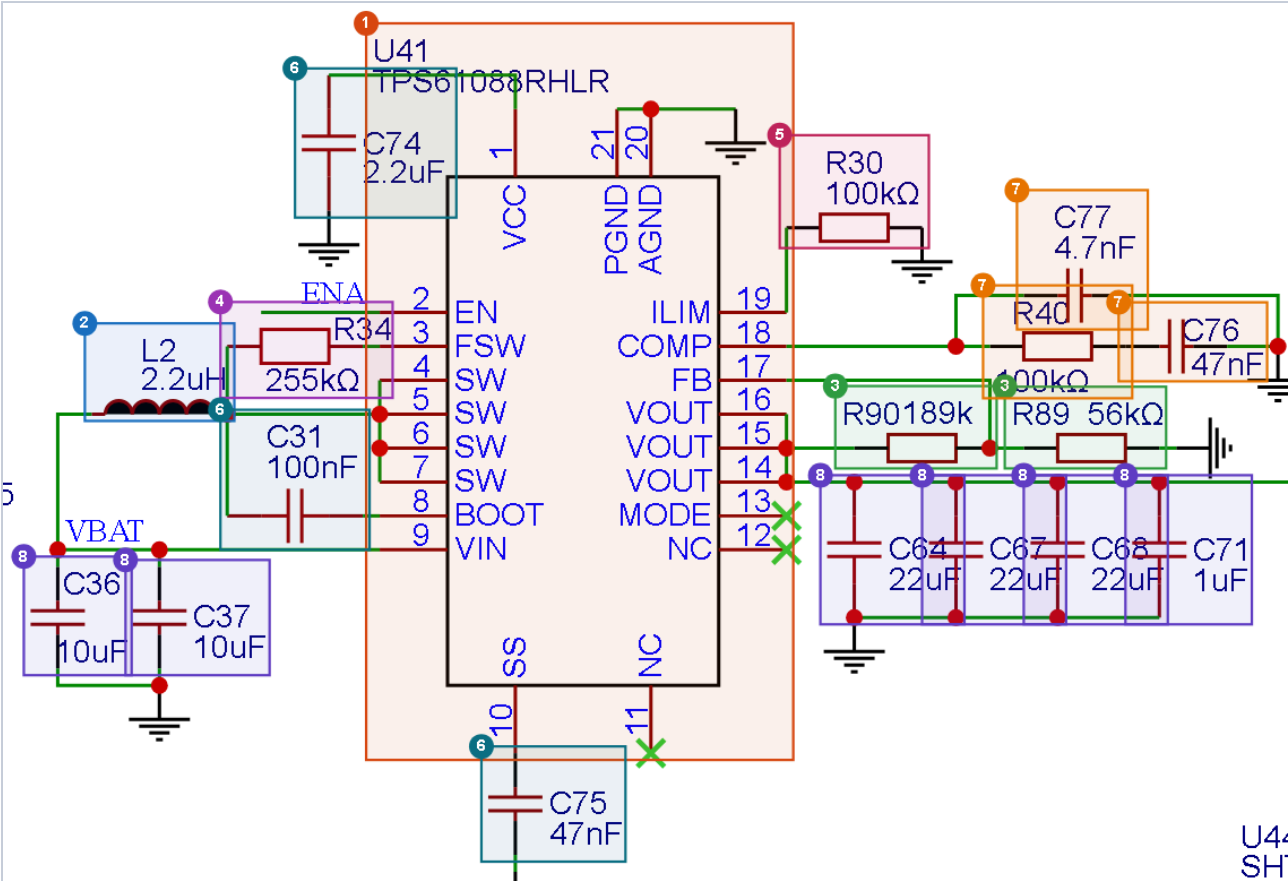


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-05	1.0	2026-07-19	Battery boost to about 24 V (U1, TPS61175)	5 of 26

Block function: Boosts the battery to about 24 V as a backup of the +24V rail. **Requirement covered:** 5 V rail / backup of the +24V rail

No	Component	What it is	What it does
1	C5, C6	22 uF and 100 nF capacitors on VBAT.	Decoupling on the load-switch output.
2	U1	Boost converter TPS61175 (VIN = VBAT).	Boosts the battery voltage to about 24 V. Enabled by the EN_BOOST signal from the ESP32 (IO15).
3	L5, D12	10 uH inductor and SK36 Schottky diode.	Boost inductor and rectifier diode from the SW node to the boost output.
4	D13	SS54 Schottky diode from the boost output to +24V.	ORs the boost output onto the +24V rail without back-feeding into the boost converter.
5	R3, R2	FB divider 187k / 10k.	Sets the output to about $1.229 \text{ V} \times (1 + 187/10) = 24 \text{ V}$ (calculated from the resistor values).
6	R4, R5	100k pull-down and 1k series resistor on EN.	Keep the boost off while EN_BOOST is inactive; the 1k limits current from the GPIO.
7	R6, R7, C7, C8	82k FREQ resistor, 10k/22 nF COMP network, 47 nF soft-start.	Set the switching frequency, loop compensation and soft-start.
8	C9, C10	Output capacitors 22 uF/35 V and 10 uF/50 V.	Boost output filter.

Figure 6 - Battery boost to 5 V (U41, TPS61088)



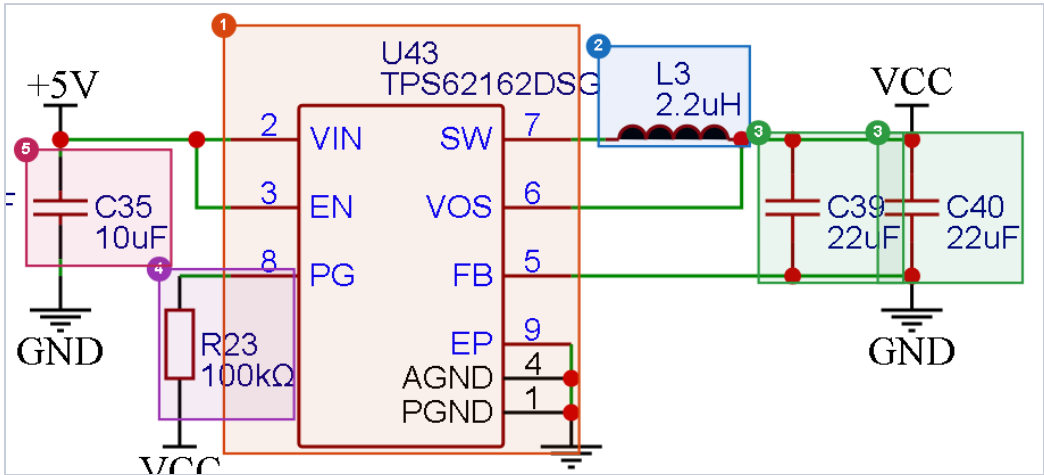
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-06	1.0	2026-07-19	Battery boost to 5 V (U41, TPS61088)	6 of 26

Block function: Boosts the battery to 5VBOOST. **Requirement covered:** 5 V rail

No	Component	What it is	What it does
1	U41	Boost converter TPS61088 (VIN = VBAT, VOUT = 5VBOOST, EN = ENA).	Generates 5 V from the battery. Active when the ENAsignal from the power-latch circuit is high.
2	L2	2.2 uH inductor.	Boost inductor between VBAT and the SW node.
3	R90, R89	FB divider 189k / 56k.	Sets the output to about 1.204 V x (1 + 189/56) = 5.3 V (calculated from the resistor values).
4	R34	255k resistor on FSW.	Sets the switching frequency.
5	R30	100k resistor on ILIM.	Sets the peak current limit.
6	C74, C75, C31	Internal VCC capacitor, soft-start capacitor and bootstrap capacitor.	Support parts for the IC.
7	R40, C76, C77	COMP compensation network.	Stabilizes the boost control loop.
8	C64, C67, C68, C71, C36, C37	VBAT input and 5VBOOST output capacitors.	Decoupling on the input and output sides of the boost.

3.3 V rail

Figure 7 - 5 V to 3.3 V buck (U43)

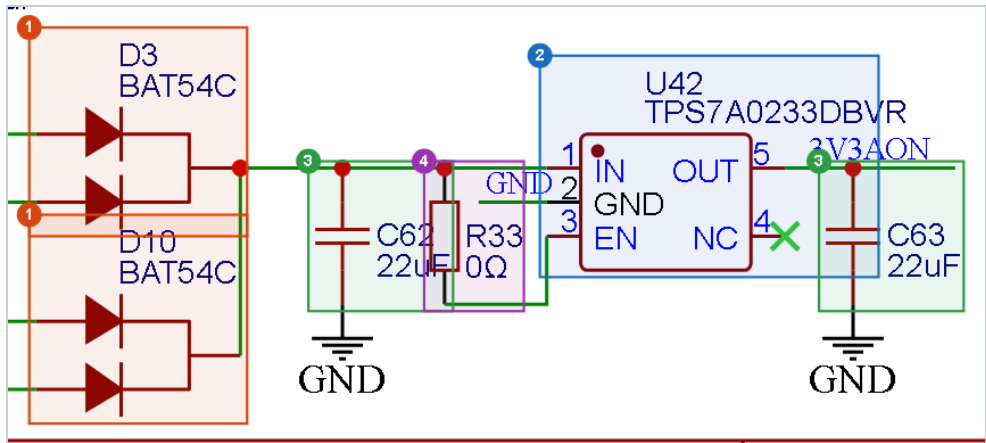


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-07	1.0	2026-07-19	5 V to 3.3 V buck (U43)	7 of 26

Block function: Buck that generates the 3.3 V VCC rail. **Requirement covered:** 3.3 V rail

No	Component	What it is	What it does
1	U43	Buck TPS62162 (VIN = +5V, EN = +5V, VOS = VCC).	Steps +5V down to 3.3 V (fixed-voltage version, FB to GND). Always on because EN is tied to +5V.
2	L3	2.2 uH inductor.	Buck inductor from the SW node to VCC.
3	C39, C40	10 uF / 22 uF capacitors on VCC.	Bulk capacitors of the 3.3 V output.
4	R23	100k pull-up on PG.	PG is not connected to any other component in this netlist (pull-up only).
5	C35	+5V capacitor, PG pull-up and VCC output capacitors.	Decoupling of the input/output of the 3.3 V buck; R23 is the pull-up on the PG pin.

Figure 8 - Always-on 3.3 V LDO and its source selection (U42)

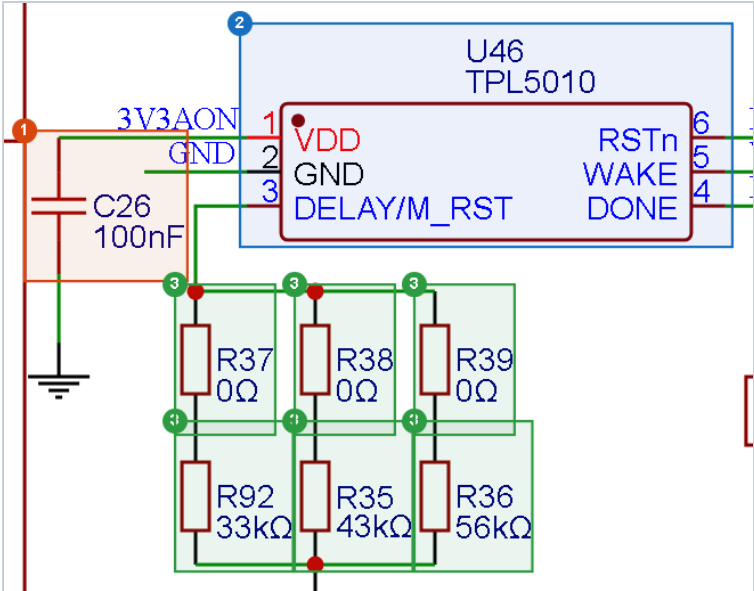


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-08	1.0	2026-07-19	Always-on 3.3 V LDO and its source selection (U42)	8 of 26

Block function: The 3V3AON LDO and its source selection. Requirement covered: 3.3 V rail

No	Component	What it is	What it does
1	D3, D10	Dual diode BAT54C (common cathode).	D3 ORs VBAT_IN and 5VBUCK; D10 ORs 5VEXT and VBUS (USB). The result is the net that feeds the LDO input.
2	U42	LDO TPS7A0233 (3.3 V, very low quiescent current).	Converts the available source into 3V3AON. EN is pulled to the input through R33 (0 ohm), so it is always enabled.
3	C62, C63	22 uF and 100 nF input/output capacitors.	Decoupling of the LDO and the 3V3AON rail.
4	R33	0 ohm resistor between the input and the EN pin of U42.	Ties EN to the input so the LDO is always enabled (can be removed if a separate EN is needed).

Figure 9 - TPL5010 timer/watchdog (U46)

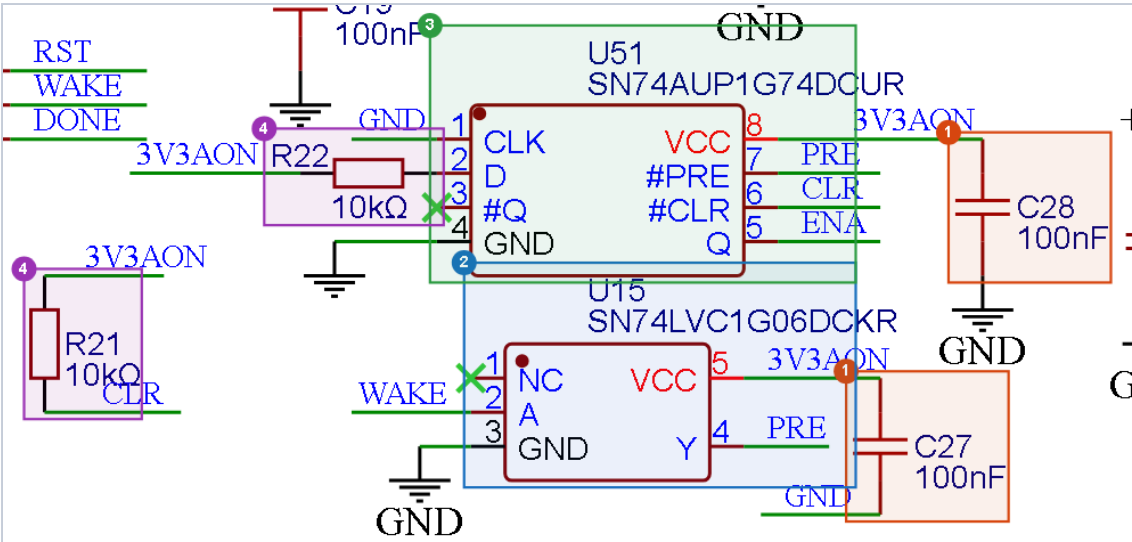


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-09	1.0	2026-07-19	TPL5010 timer/watchdog (U46)	9 of 26

Block function: Nano-power timer/watchdog and its interval selector. **Requirement covered:** 3.3 V rail / power latch

No	Component	What it is	What it does
1	C26	22 uF and 100 nF input/output capacitors.	Decoupling of the LDO and the 3V3AON rail.
2	U46	Nano-power timer TPL5010.	Generates periodic WAKE pulses and works as a watchdog: DONE from the ESP32 (IO17) signals that the firmware is still alive, RSTn goes to RST (ESP32 reset).
3	R35, R36, R92, R37, R38, R39	43k / 56k / 33k resistors to GND with three 0 ohm jumpers.	Select the TPL5010 timer interval through the DELAY/M_RST pin (one 0 ohm jumper is fitted for the chosen interval).

Figure 10 - ENAlatch (U51) and open-drain inverter (U15)

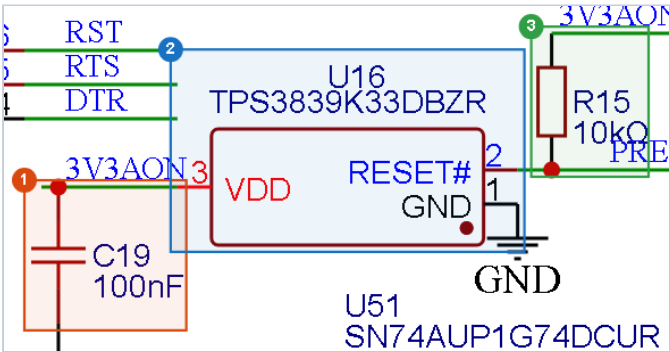


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-10	1.0	2026-07-19	ENAlatch (U51) and open-drain inverter (U15)	10 of 26

Block function: ENAlatch flip-flop and open-drain inverter. Requirement covered: 3.3 V rail / power latch

No	Component	What it is	What it does
1	C27, C28	22 uF and 100 nF input/output capacitors.	Decoupling of the LDO and the 3V3AON rail.
2	U15	Open-drain inverter SN74LVC1G06.	Turns WAKE from U46 into a pull-down on net PRE (R15 pull-up to 3V3AON).
3	U51	D flip-flop SN74AUP1G74 (CLK to GND, Q = ENA).	Used as a latch: PRE low sets ENA, CLR low from the ESP32 (IO16) clears ENA. ENA controls U14 (load switch) and U41 (5 V boost).
4	R21, R22	10k pull-ups to 3V3AON on PRE, CLR and the flip-flop D input.	Keep the default logic levels.

Figure 11 - 3V3AON voltage supervisor (U16)



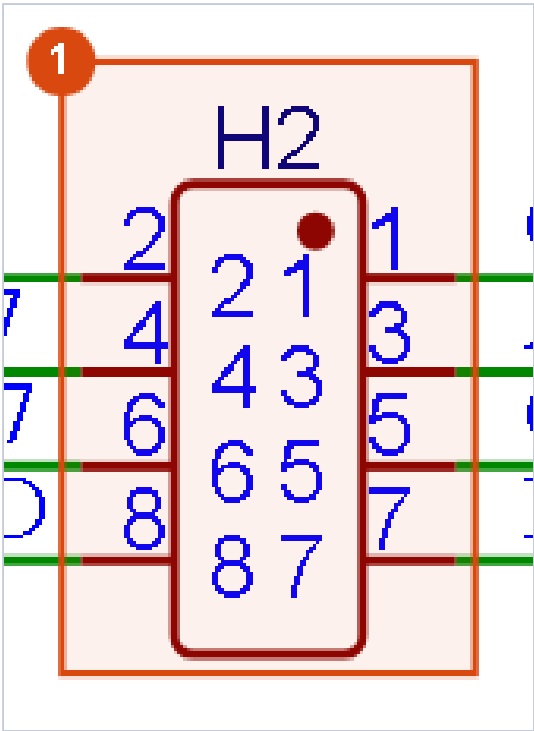
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-11	1.0	2026-07-19	3V3AON voltage supervisor (U16)	11 of 26

Block function: Voltage supervisor of 3V3AON. Requirement covered: 3.3 V rail / power latch

No	Component	What it is	What it does
1	C19	22 uF and 100 nF input/output capacitors.	Decoupling of the LDO and the 3V3AON rail.
2	U16	Voltage supervisor TPS3839K33 (threshold variant for a 3.3 V rail).	RESET# output is connected to net PRE; supervises 3V3AON.
3	R15	10k pull-ups to 3V3AON on PRE, CLR and the flip-flop D input.	Keep the default logic levels.

MQ heater supply

Figure 12 - Sensor module connector H2 (one connector as example)

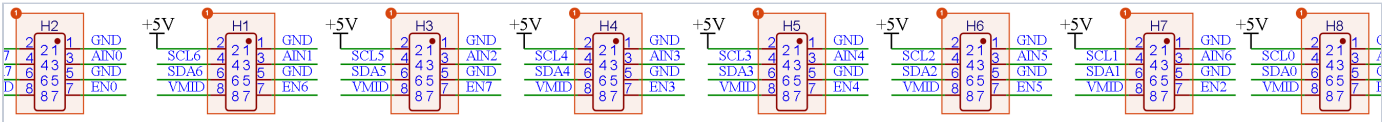


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-12	1.0	2026-07-19	Sensor module connector H2 (one connector as example)	12 of 26

Block function: One sensor module connector as an example of every pin function. **Requirement covered:** MQ heater supply and MQ sensor interface

No	Component	What it is	What it does
1	H2	Sensor module connectors (H1..H8).	Each connector carries: pin 2 = +5V (module/heater supply), pin 8 = VMID (bias), pin 3 = AINx (analog signal to the ADC), pins 4/6 = SCLx/SDAx (I2C channel from the mux), pin 7 = ENx (module enable), pins 1/5 = GND. The heater switch and sensor load resistor are not on this board (assumed to be on the sensor module; to be confirmed).

Figure 13 - Sensor module connectors H1-H8

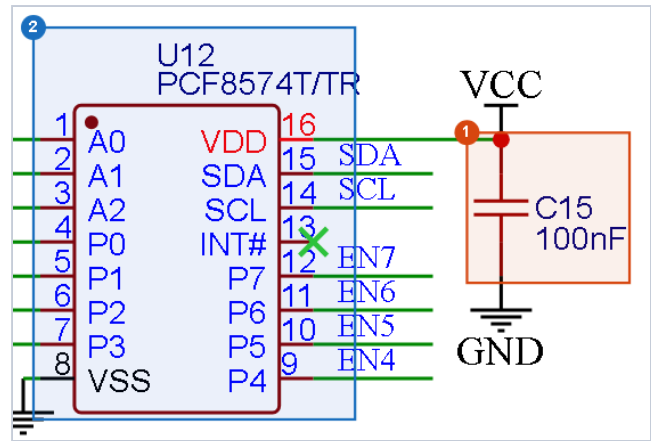


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-13	1.0	2026-07-19	Sensor module connectors H1-H8	13 of 26

Block function: The eight sensor module connectors H1-H8. **Requirement covered:** MQ heater supply and MQ sensor interface

No	Component	What it is	What it does
1	H1, H2, H3, H4, H5, H6, H7, H8	Sensor module connectors (H1..H8).	Each connector carries: pin 2 = +5V (module/heater supply), pin 8 = VMID (bias), pin 3 = AINx (analog signal to the ADC), pins 4/6 = SCLx/SDAx (I2C channel from the mux), pin 7 = ENx (module enable), pins 1/5 = GND. The heater switch and sensor load resistor are not on this board (assumed to be on the sensor module; to be confirmed).

Figure 14 - Sensor module enable I/O expander (U12)



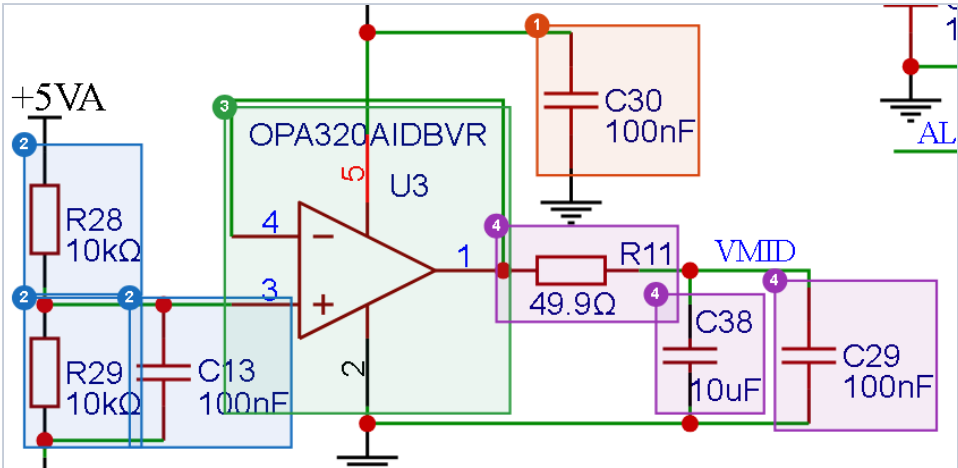
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-14	1.0	2026-07-19	Sensor module enable I/O expander (U12)	14 of 26

Block function: I/O expander that enables the sensor modules. **Requirement covered:** MQ heater supply (module enable)

No	Component	What it is	What it does
1	C15	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
2	U12	I/O expander PCF8574.	Outputs P0-P7 become EN0-EN7 to enable the sensor modules one by one.

MQ sensor bridge / analog circuitry

Figure 15 - VMID bias of about 2.5 V (U3)

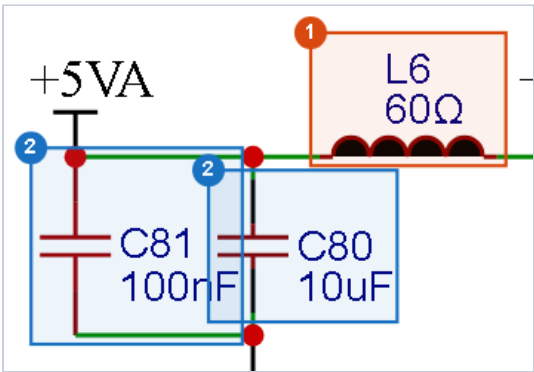


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-15	1.0	2026-07-19	VMID bias of about 2.5 V (U3)	15 of 26

Block function: Generator of the VMID bias of about 2.5 V. Requirement covered: MQ sensor bridge / analog circuitry

No	Component	What it is	What it does
1	C30	10 uF and 100 nF capacitors on +5VA.	Decoupling for the ADC, op-amps and reference.
2	R28, R29, C13	10k / 10k divider and 100 nF.	Creates half of +5VA(about 2.5 V).
3	U3	Op-amp OPA320 as a buffer.	Buffers the mid-voltage into a low-impedance VMID source.
4	R11, C29, C38	49.9 ohm resistor and capacitors.	Isolate the buffer output and keep it stable with capacitive loads; VMID goes to pin 8 of every sensor connector.

Figure 16 - Analog +5VAsupply filter (L6)

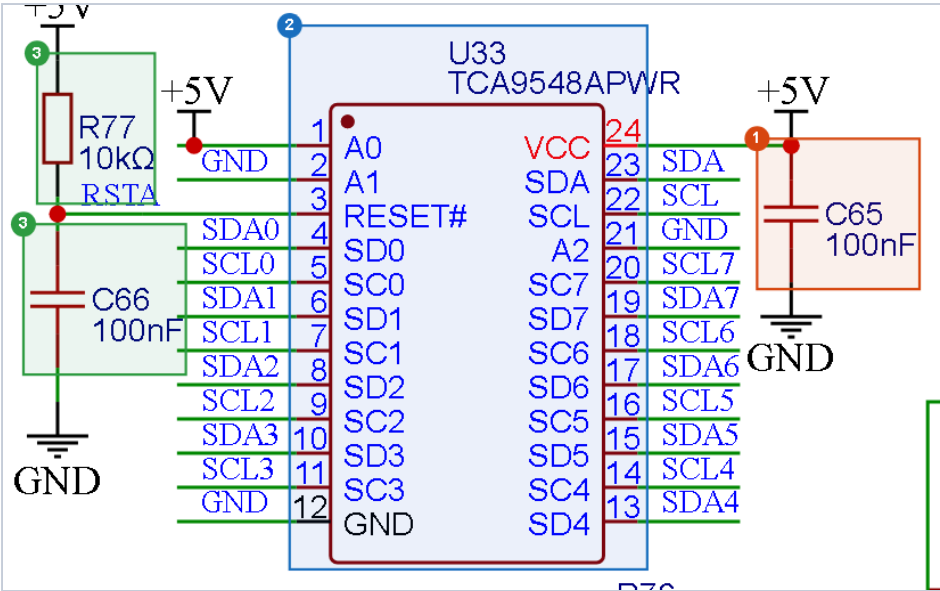


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-16	1.0	2026-07-19	Analog +5VAsupply filter (L6)	16 of 26

Block function: Analog +5VAsupply filter. Requirement covered: MQ sensor bridge / analog circuitry

No	Component	What it is	What it does
1	L6	60 ohm ferrite bead.	Separates digital noise from the analog +5VA rail.
2	C80, C81	10 uF and 100 nF capacitors on +5VA.	Decoupling for the ADC, op-amps and reference.

Figure 17 - Sensor module I2C multiplexer (U33)

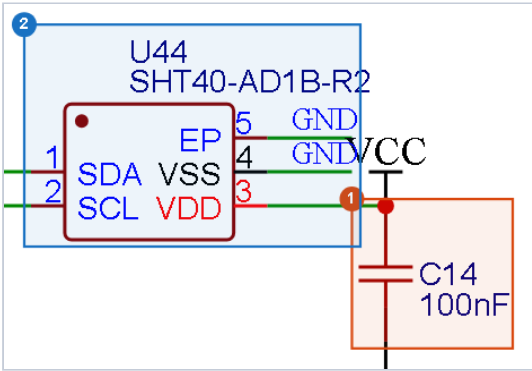


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-17	1.0	2026-07-19	Sensor module I2C multiplexer (U33)	17 of 26

Block function: I2C multiplexer for the eight sensor modules. **Requirement covered:** MQ sensor bridge / analog circuitry

No	Component	What it is	What it does
1	C65	10 uF and 100 nF capacitors on +5V.	Decoupling of the +5V rail.
2	U33	I2C multiplexer TCA9548A(8 channels).	Splits the main I2C bus into eight channels SDA0-7/SCL0-7 toward the modules; RESET# is held high by R77.
3	R77, C66	10k pull-up and 100 nF on RSTA.	Keep the mux from being reset unless intended.

Figure 18 - Temperature and humidity sensor (U44)



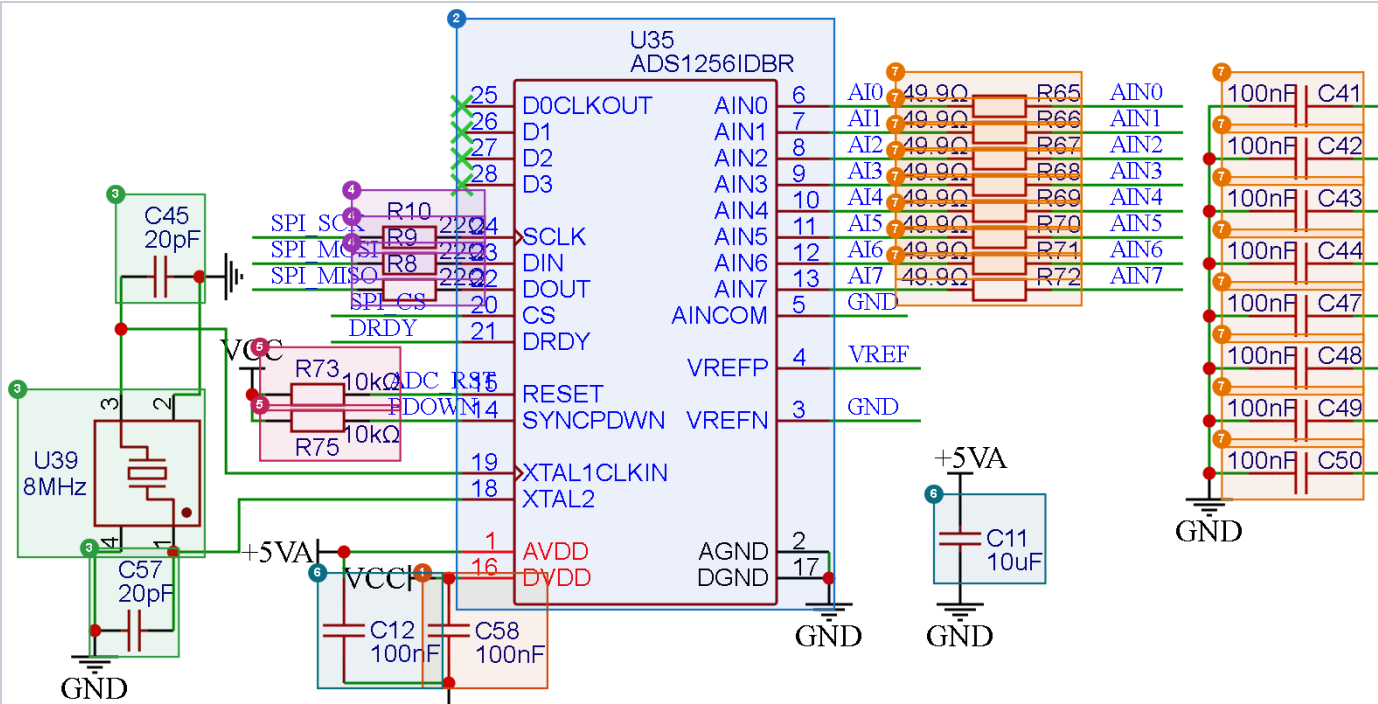
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-18	1.0	2026-07-19	Temperature and humidity sensor (U44)	18 of 26

Block function: Environment sensor inside the board. **Requirement covered:** MQ sensor bridge / analog circuitry (environment compensation)

No	Component	What it is	What it does
1	C14	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
2	U44	Temperature/humidity sensor SHT40.	Measures temperature and humidity inside the board on the main I2C bus.

ADC

Figure 19 - ADS1256 ADC with input filters (AIN), clock and decoupling

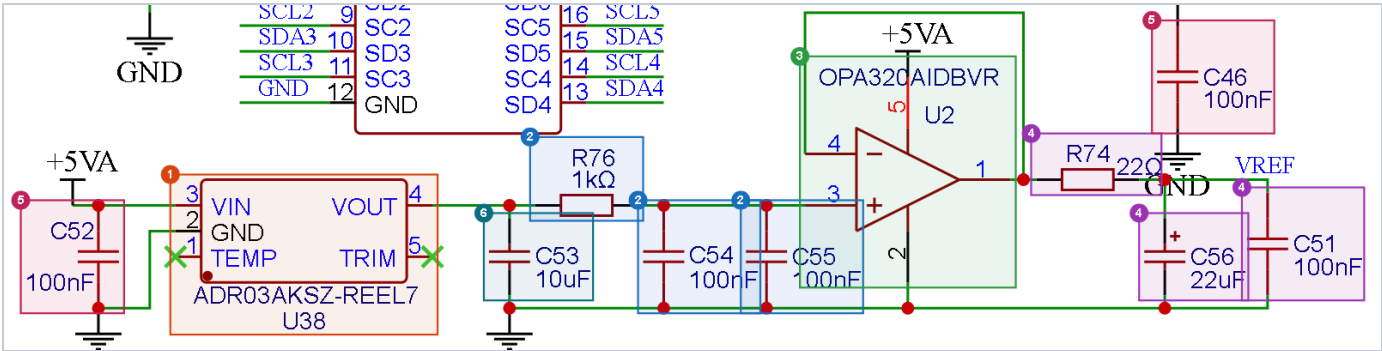


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-19	1.0	2026-07-19	ADS1256 ADC with input filters (AIN), clock and decoupling	19 of 26

Block function: ADS1256 with its input filters, clock and decoupling. **Requirement covered:** ADC; MQ sensor bridge / analog circuitry

No	Component	What it is	What it does
1	C58	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
2	U35	Delta-sigma ADC ADS1256 (8 channels, 24 bit).	Converts the AI0-AI7 voltages to digital data. AVDD = +5VA, DVDD = VCC, VREFP = VREF, VREFN and AINCOM = GND. Controlled over SPI (CS, SCLK, DIN, DOUT) + DRDY.
3	U39, C45, C57	4-pin SMD crystal with two 20 pF capacitors.	ADC clock source (8 MHz according to the U39 label on the schematic).
4	R8, R9, R10	22 ohm series resistors on MISO, MOSI, SCK.	Damp ringing/EMI on the SPI lines to the ADC.
5	R73, R75	10k pull-ups on RESET and SYNC/PDWN.	Keep the ADC active by default; the ESP32 can pull them low (ADC_RST, PDOWN).
6	C11, C12	10 uF and 100 nF capacitors on +5VA.	Decoupling for the ADC, op-amps and reference.
7	R65, R66, R67, R68, R69, R70, R71, R72, C41, C42, C43, C44, C47, C48, C49, C50	49.9 ohm series resistor and 100 nF capacitor to GND on every channel.	About 32 kHz low-pass (calculated) that suppresses noise and protects the ADC inputs; AINx from the connector becomes Aix at the ADS1256.

Figure 20 - 2.5 V voltage reference and VREF buffer



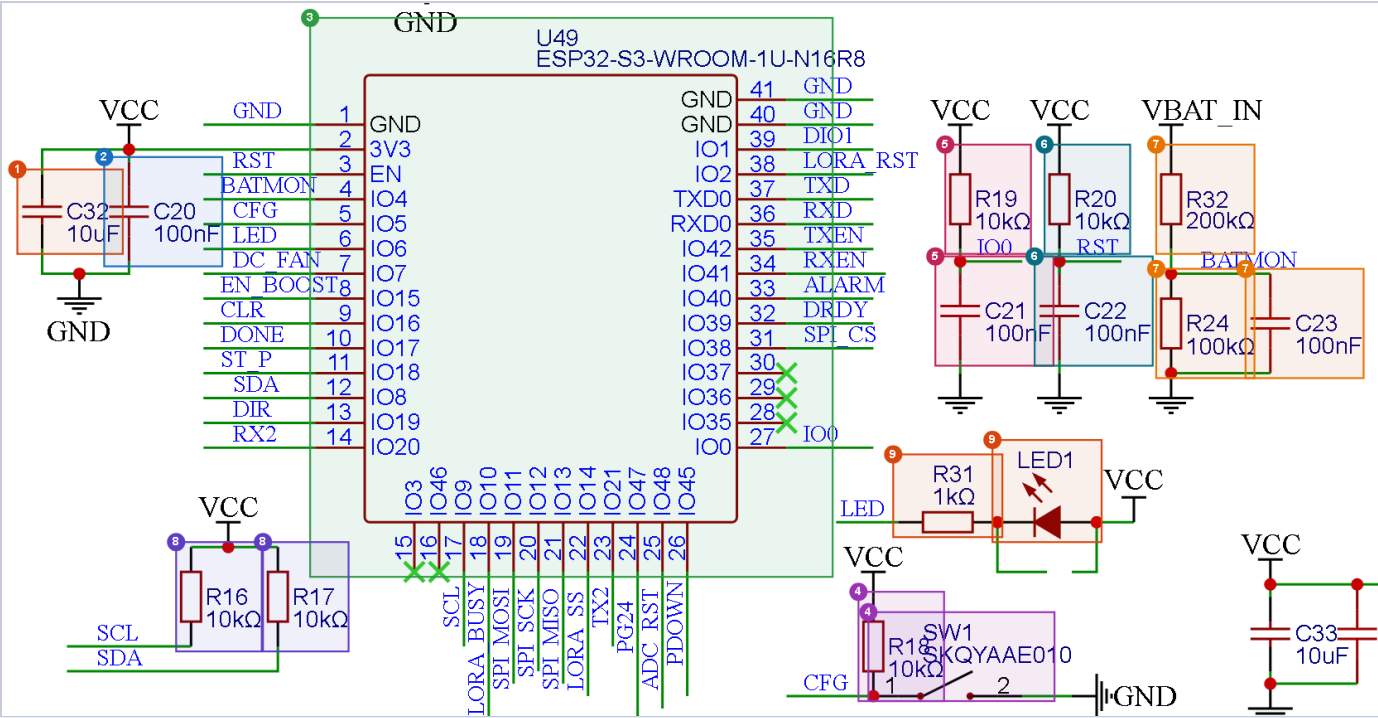
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-20	1.0	2026-07-19	2.5 V voltage reference and VREF buffer	20 of 26

Block function: 2.5 V reference and VREF buffer. Requirement covered: ADC

No	Component	What it is	What it does
1	U38	Voltage reference ADR03 (2.5 V).	Precision VREF source for the ADC (VIN = +5VA).
2	R76, C54, C55	1k resistor and 2x 100 nF.	Low-pass filter of the reference noise before the buffer.
3	U2	Op-amp OPA320 as a buffer (voltage follower).	Buffers VREF so it is stiff against the load and the switching current of the ADC.
4	R74, C56, C51	22 ohm resistor, 22 uF tantalum, 100 nF.	Isolation and reservoir capacitors on the ADC VREF pin.
5	C46, C52	10 uF and 100 nF capacitors on +5VA.	Decoupling for the ADC, op-amps and reference.
6	C53	10 uF capacitor on VREF_2V5 (output of U38).	Reservoir/filter of the 2.5 V reference output.

ESP32-S3

Figure 21 - ESP32-S3-WROOM-1U-N16R8 with support parts (including status LED)



Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-21	1.0	2026-07-19	ESP32-S3-WROOM-1U-N16R8 with support parts (including status LED)	21 of 26

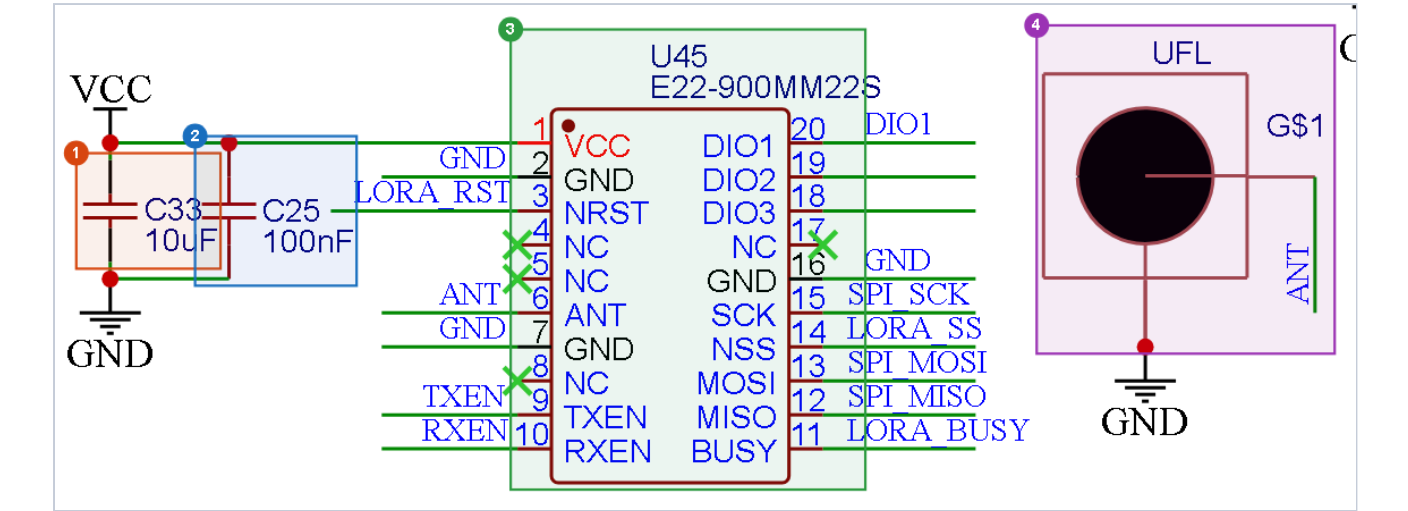
Block function: Main microcontroller with reset, boot, LED and I2C pull-ups. **Requirement covered:** ESP32-S3; LED / buzzer driver (LED)

No	Component	What it is	What it does
1	C32	10 uF / 22 uF capacitors on VCC.	Bulk capacitors of the 3.3 V output.
2	C20	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
3	U49	Module ESP32-S3-WROOM-1U-N16R8 (external antenna, 16 MB flash, 8 MB PSRAM).	The brain of the GLD: reads the ADC (SPI_CS/DRDY/ADC_RST/PDOWN), controls the LoRa radio (LORA_SS/RST/BUSY/DIO1/TXEN/RXEN), I2C (SDA/SCL), RS-485 (TX2/RX2/DIR), alarm, fan, LED and boost, and reads BATMON, PG24, ST_P and CFG.
4	SW1, R18	CFG button with a 10k pull-up.	Configuration button: pulls CFG to GND when pressed.
5	R19, C21	10k pull-up and 100 nF on IO0.	Keep IO0 high (normal boot); Q1 pulls it low in download mode.
6	R20, C22	10k pull-up and 100 nF on EN/RST.	RC reset circuit for the ESP32; also driven by the TPL5010.
7	R32, R24, C23	200k / 100k divider to BATMON with 100 nF.	Scales VBAT_IN down so the ESP32 ADC (IO4) can read it safely; the 1/3 ratio matches the firmware formula 'avg x 3'.
8	R16, R17	10k I2C pull-ups to VCC.	Pull-ups for SDA and SCL.
9	LED1, R31	0603 LED (19-217/BHC-ZL1M2RY/3T) with a 1k resistor.	Lights when the ESP32 pulls the LED pin (IO6) to GND; anode to VCC (active-low).

Block function: USB connector, ESD protection, CH340C and auto-reset. **Requirement covered:** ESP32-S3 (programming)

No	Component	What it is	What it does
1	C34	10 uF / 22 uF capacitors on VCC.	Bulk capacitors of the 3.3 V output.
2	C24	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
3	USB1	Micro-USB connector.	Programming and serial port to the PC; VBUS also feeds the 3V3AON source OR-ing (3.3 V rail).
4	D4, D5, D11	ESD protection on D-, D+ and VBUS.	Protects the USB lines from electrostatic discharge.
5	U48	USB-UART bridge CH340C (VCC 3.3 V).	Translates USB to the ESP32 UART0 (TXD/RXD).
6	Q1	Dual NPN with resistors, UMH3N.	Classic auto-reset circuit: DTR/RTS from the CH340 drive IO0 and RST of the ESP32 during upload.
7	R96	470 ohm resistor.	Series resistor on the TXD line from the ESP32 to RXD of the CH340C.

Figure 23 - E22-900MM22S LoRa module and antenna connector



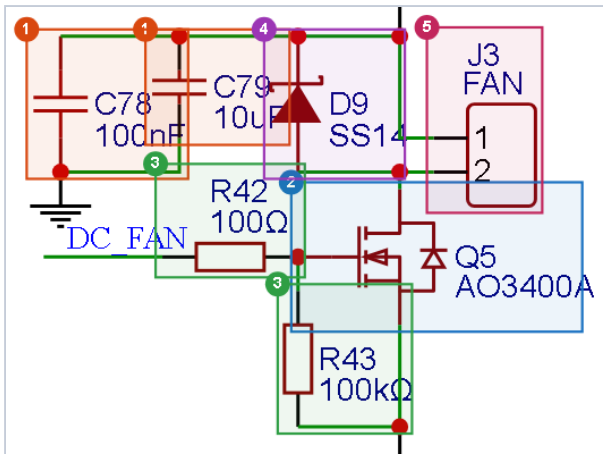
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-23	1.0	2026-07-19	E22-900MM22S LoRa module and antenna connector	23 of 26

Block function: LoRa module and antenna connector. Requirement covered: LoRa

No	Component	What it is	What it does
1	C33	10 uF / 22 uF capacitors on VCC.	Bulk capacitors of the 3.3 V output.
2	C25	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
3	U45	LoRa module E22-900MM22S (SX126x class, 900 MHz).	Interface: SPI (SPI_MISO/MOSI/SCK + LORA_SS), LORA_RST, LORA_BUSY, DIO1, and TXEN/RXEN for the TX/RX switch. Supplied from VCC (3.3 V).
4	G1	U.FL antenna connector.	Connection point of the antenna cable to the ANT pin of the LoRa module.

Fan driver

Figure 24 - 5 V fan driver (Q5, J3)



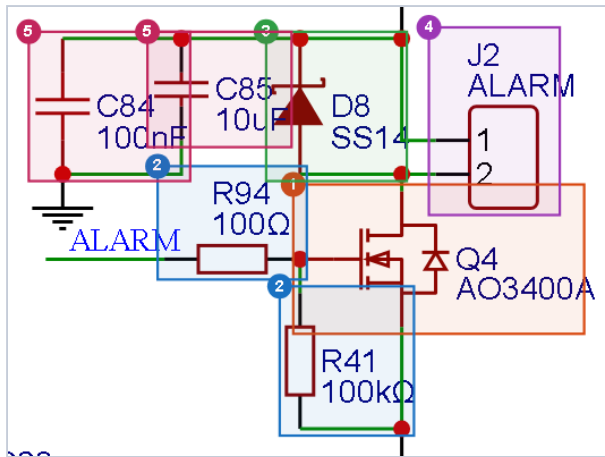
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-24	1.0	2026-07-19	5 V fan driver (Q5, J3)	24 of 26

Block function: 5 V fan driver. Requirement covered: Fan driver

No	Component	What it is	What it does
1	C79, C78	10 uF and 100 nF capacitors on +5V.	Decoupling of the +5V rail.
2	Q5	N-MOSFET AO3400Aas a low-side switch.	Turns on the fan at J3 when DC_FAN is high.
3	R42, R43	100 ohm gate resistor and 100k pull-down.	Gate current limiter and default-off holder.
4	D9	SS14 flyback diode.	Clamps the inductive spike of the fan to +5V.
5	J3	Fan connector (pin 1 = +5V, pin 2 = Q5 drain).	Connection point of the 5 V fan.

LED / buzzer driver

Figure 25 - 24 V alarm output driver (Q4, J2)



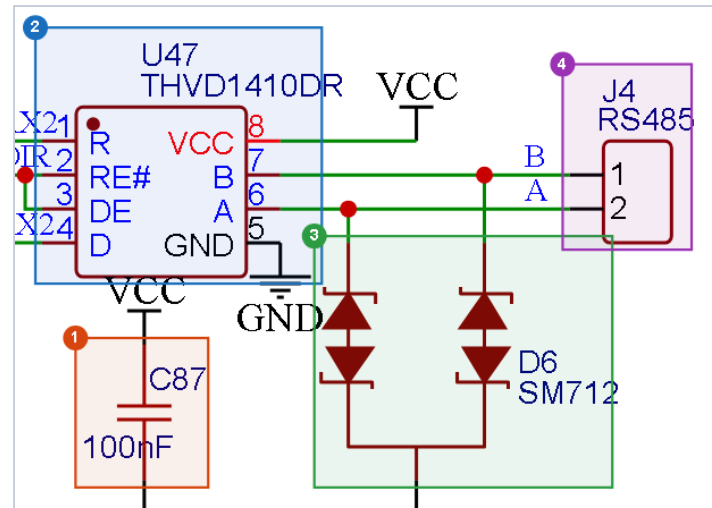
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-25	1.0	2026-07-19	24 V alarm output driver (Q4, J2)	25 of 26

Block function: 24 V alarm output driver. Requirement covered: LED / buzzer driver

No	Component	What it is	What it does
1	Q4	N-MOSFET AO3400A as a low-side switch.	Turns on the load at J2 when ALARM is high; its source goes to GND.
2	R94, R41	100 ohm gate resistor and 100k pull-down.	R94 limits the gate charging current; R41 keeps Q4 off while the GPIO is not yet active.
3	D8	SS14 flyback diode.	Clamps the inductive spike of the load at J2 to +24V.
4	J2	Alarm load connector (pin 1 = +24V, pin 2 = Q4 drain).	Connection point of the 24 V alarm load (load type to be confirmed).
5	C84, C85	100 nF and 10 uF capacitors on +24V.	Decoupling of the +24V rail next to the alarm driver.

RS-485 / Modbus

Figure 26 - RS-485 transceiver (U47)



Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-26	1.0	2026-07-19	RS-485 transceiver (U47)	26 of 26

Block function: RS-485 transceiver, TVS protection and connector. Requirement covered: RS-485 / Modbus

No	Component	What it is	What it does
1	C87	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
2	U47	RS-485 transceiver THVD1410 (3.3 V).	Converts the TX2/RX2 UART of the ESP32 into the differential A/B signal; direction is controlled by DIR (drives RE# and DE together).
3	D6	TVS SM712 (asymmetric, made for RS-485).	Protects pins A/B against surges and ESD.
4	J4	RS-485 connector (A, B).	Bus connection terminal. There is no 120 ohm termination or bias resistor on this schematic.

Protection devices

Designator	Part	Location	Protects against / function	Figure
F1	MINISMDC260F/16-2	In series with 24V+ after J1	Overcurrent on the 24 V line (PTC)	1
D1	SMBJ33A	After F1, to 24V-	Voltage surges on the 24 V line (TVS)	1
L1	ACM7060-301-2PL	In series with 24V+ and return	Common-mode noise	1
L4	MPZ2012S101AT000	In series with +24V	High-frequency noise (ferrite)	1
D2 / R26	LBZT52C12T1G / 100k	Gate of Q3	12 V Vgs limit for Q3	1
Q3	SI7465DP-T1-GE3	+24V to buck input	High-side switch (orientation to be verified)	1
F2	MINISMDC260F/16-2	In series with battery input J1.2	Battery overcurrent (PTC)	2
Q2 / R1	AO4407 / 100k	Battery to VBAT_IN	High-side switch (orientation to be verified)	2
D7 / D13	SS54	5VEXT to +5V; boost to +24V	Reverse-current blocking (OR-ing)	2, 5
D3 / D10	BAT54C	Source of U42 (3V3AON)	OR-ing of battery / 5VBUCK / 5VEXT / VBUS	8
D8 / D9	SS14	Loads J2 and J3	Flyback for inductive loads	24, 25
D4 / D5 / D11	LESD5D5.0CT1G	D-, D+, VBUS	USB ESD	22
D6	SM712	RS-485 A/B	RS-485 line TVS	26
L6	GZ1005D600TF	+5V to +5VA	Noise isolation of the analog supply	16

Grounding

- The whole circuit uses a single GND net. There is no separate AGND or chassis ground net in this netlist; the AGND/PGND/DGND pins of all ICs go into net GND.
- The 24 V return (J1 pin 5, 24V-) enters GND through a winding of common-mode choke L1 (pin 2 to pin 3). J1 pins 1 and 3 also go directly to GND.
- The analog supply +5VA is separated from +5V by ferrite bead L6; analog and digital ground are joined at GND.
- The shell/EP pads of USB1 and the GND pads of the module/antenna connector (G1) are connected to GND.
- A single connection point to chassis ground or PE is not drawn on the source schematic (to be confirmed against the enclosure/ATEX design).

Related figures: 1, 2, 16, 22, 23.

Notes for verification

- PTC rating of F1/F2: the part number is MINISMDC260F/16-2; the /16 suffix usually means a 16 V rating, while F1 sits on the 24 V line. Check the datasheet.
- Orientation of Q3 and Q2: the source is on the input side (+24V / battery) and the gate goes to GND through 100k. With this orientation the body diode does not appear to block reverse polarity. If reverse-polarity protection is intended, this needs to be checked.
- MQ heater supply: the board only provides +5V (pin 2) and VMID (pin 8) on each connector H1-H8. The per-channel heater switch and the sensor load resistor are not on this schematic; they are assumed to be on the sensor module (to be confirmed).
- RS-485: there is no 120 ohm termination resistor or A/B bias on the schematic, only TVS D6.
- Grounding: there is only one GND net; there is no separate AGND or chassis ground. The connection point to chassis ground/PE is not drawn.
- The type of the 24 V alarm load on J2 cannot be determined from the schematic.
- The converter output voltages quoted (about 5.0 V, 24 V, 5.3 V) are calculated from the feedback resistors and the typical reference voltage of the IC; they have not been measured.
- The figures are rendered from the original schematic file (same appearance as EasyEDA) so that the boundary boxes sit exactly on the components.